# Markus Tatzgern

*Curriculum Vitae*




Email mail@markustatzgern.com
Web markustatzgern.com
Scholar scholar.google.com/citations?user=Fs0fyasAAAAJ
ORCID 0000-0002-3900-4944


## Profile

My research develops visualization and interaction techniques for Augmented and Virtual Reality that bridge the digital and the physical world. I explore adaptive visualization, multimodal interaction and AI-driven approaches for human-centered XR, with applications in training and learning, healthcare, human–robot interaction and everyday augmentation.

| | |
|---|---|
| Publications | 7 journal and 26 conference papers; 18 at CORE A* venues (e.g. ACM CHI, ACM UIST, IEEE ISMAR, IEEE VR), 6 in SJR Q1 journals; 4 best paper honorable mentions |
| Funding | More than € 3.4m acquired for my lab since 2015; involved in proposals totalling about € 8.7m |
| Leadership | Science & Technology Program Chair IEEE ISMAR 2024, Papers Chair IEEE ISMAR 2025; Associate Editor IEEE TVCG; Head of Research, Salzburg University of Applied Sciences (2023–2026) |

## Employment

| | |
|---|---|
| since 09/2026 | **Associate Professor of Creative Technologies**, Warwick Manufacturing Group & School of Cross-Faculty Studies, University of Warwick, England |
| 01/2023–08/2026 | **Head of Research**, Department of Creative Technologies, Salzburg University of Applied Sciences, Austria |
| 06/2018–08/2026 | **FH-Professor**, Mixed Reality and Game Development, Salzburg University of Applied Sciences, Austria |
| 09/2015–08/2026 | **Senior Lecturer, Head of Academic Area**, Game & Mixed Reality, Salzburg University of Applied Sciences, Austria |
| 09/2009–08/2015 | **Research Assistant**, Christian Doppler Laboratory for Handheld Augmented Reality, Institute of Computer Graphics and Vision, Graz University of Technology, Austria |

## Education

2015 **Dr.techn. in Computer Science** (Doktor der technischen Wissenschaften, equivalent to PhD)
Graz University of Technology, Institute of Computer Graphics and Vision, Austria
Thesis: *Situated Visualization in Augmented Reality*
Graduated with highest distinction

2008 **Dipl.-Ing. in Telematics** (Diplom-Ingenieur, equivalent to MSc)
Graz University of Technology, Institute of Computer Graphics and Vision, Austria
Thesis: *A Framework for Automatically Creating 3D Explosion Diagrams*
Graduated with highest distinction

2005 **Bakk.techn. in Telematics** (Bakkalaureus der technischen Wissenschaften, equivalent to BSc)
Graz University of Technology, Austria

## Research Funding

In total, I have acquired more than € 3.4m for my lab since joining the Salzburg University of Applied Sciences in 2015 and have participated in project proposals that resulted in about € 8.7m of funding. In the following, the total project budget and the university share are given as 'total (share)'. Contract research for industry partners is not listed due to data protection and non-disclosure agreements.

### Salzburg University of Applied Sciences

05/2025–04/2029 **GenAIedTech - Content Creation with Generative Artificial Intelligence for Interactive Educational Technologies**
Funded by the **FFG** Austrian Research Promotion Agency / COIN Aufbau
Budget: 720.000€
Role: Principal investigator, project acquisition

04/2025–06/2027 **Extended Reality and Micro Learning for Vocational Training.**
Funded by the **Republic of Austria**, Federal Ministry Agriculture and Forestry, Climate and Environmental Protection, Regions and Water Management,
Partners: Adult Education Center Burgenland, University of Salzburg
Budget: 262.000€ (75.000€)
Role: Co-investigator, project acquisition

01/2025–12/2026 **TGuard: Combating Disinformation through the Analysis and Detection of Fake News Networks**
Funded by the **FFG** Austrian Research Promotion Agency / K-PASS 2023
Partner: Austrian Institute of Technology
Budget: 735.000€ (314.000€)
Role: Project development and acquisition as Head of Research

01/2024–12/2026 **FAIR AI: Fostering Austria's Innovative Strength and Research excellence in Artificial Intelligence**
Funded by the **FFG** Austrian Research Promotion Agency / Artificial Intelligence Mission Austria (AIM AT)
Lead partner: Austrian Institute of Technology
Budget: 3.0M€ (100.000€)
Role: Co-investigator, project acquisition

05/2022–04/2025 **CIRCUIT: Towards Comprehensive CBCT Imaging Pipelines for Real-time Acquisition, Analysis, Interaction and Visualization**
Funded by the Austrian Research Promotion Agency / **FFG** Bridge (no FO999891105)
Industry partner: medPhoton GmbH
Budget: 470.000€
Role: Principal investigator, project acquisition

05/2022–04/2026 **doc.hci – Designing Meaningful Human-Technology Relations**
Funded by **FWF** Austrian Science Fund, doc.funds.connect
Research partner: University of Salzburg
Budget: 1.0M€ (417.000€)
Role: Co-investigator, project acquisition

03/2021–02/2024 **EdTech Salzburg - Augmented Learning Lab**
Funded by Federal State Government of Salzburg, Austria
Research partner: Salzburg University of Education Stefan Zweig, University of Salzburg
Budget: 1.1M€ (393.000€)
Role: Principal investigator, project acquisition

06/2020–11/2023 **Adaptive Visualization for Augmented Reality**
Funded by the Austrian Research Promotion Agency / **FFG** Bridge (no 877104), Austria
Industry partner: Wikitude GmbH / Qualcomm Ltd.
Budget: 383.000€
Role: Project acquisition, principal investigator

03/2020–08/2022 **EdTech in the University Education of Healthcare Professionals**
Funded by the Federal State Government of Salzburg, Austria
Research partner: Health Sciences / Physiotherapy of Salzburg University of Applied Sciences
Budget: 110.000€
Role: Project acquisition, principal investigator

03/2020–08/2021 **Investigating Virtual Reality Environments for the Chemistry Curriculum**
Funded by the Joachim Hertz Foundation, Germany
Research partner: Paris Lodron University of Salzburg
Budget: 18.000€ (11.000€)
Role: Project acquisition, co-investigator

12/2017–11/2020 **Ecomedicine VR Physiology Lab: Information and communication technology drives innovation for health products and services**
Funded by Federal State Government of Salzburg, Austria
Research partner: Paracelsus Medical Private University
Budget: 883.000€ (317.000€)
Role: Project acquisition, co-investigator

03/2017–02/2019 **Optimizing the Online Shopping Experience using Virtual Reality**
Funded by Federal State Government of State Salzburg, Austria
Budget: 120.000€
Role: Project acquisition, co-investigator

### Graz University of Technology

03/2010–08/2015 **Christian Doppler Laboratory for Handheld Augmented Reality**
Funded by Christian Doppler Society
Industry partner: Qualcomm Inc.
Role: Early stage researcher

09/2009–02/2010 **IPCity**, Integrated Project on Interaction and Presence in Urban Environments
Funded by European Union under contract FP-2004-IST-4-27571
Role: Early stage researcher

## Publications

Co-supervised students are marked: ° PhD student, † Master's student. Titles and DOIs link to the publication.

**Journal**

1 Gerlinde Emsenhuber°, Tobias Langlotz, Denis Kalkofen, and **Markus Tatzgern**. "See what I Mean? Mobile Eye-Perspective Rendering for Optical See-Through Head-Mounted Displays". In: *IEEE Transactions on Visualization and Computer Graphics (TVCG), Special Issue IEEE International Symposium on Mixed and Augmented Reality (ISMAR)* 31.11 (2025), pp. 9603–9613. DOI: 10.1109/TVCG.2025.3616739
**SJR Q1**, **CORE A***

2 Arno Kinzinger, Winfried J. Steiner, **Markus Tatzgern**, and Christine Vallaster. "Interactive product presentation in an immersive environment: The influence of functional control on hedonic aspects". In: *Journal of Retailing and Consumer Services* 83 (2025), p. 104156. DOI: 10.1016/j.jretconser.2024.104156
**SJR Q1**

3 Arno Kinzinger, Winfried Steiner, **Markus Tatzgern**, and Christine Vallaster. "Comparing low sensory enabling (LSE) and high sensory enabling (HSE) virtual product presentation modes in e-commerce". In: *Information Systems Journal* 32.5 (2022), pp. 1034–1063. DOI: 10.1111/isj.12382
**SJR Q1**

4 Maria CR Harrington, **Markus Tatzgern**, Tom Langer†, and John W Wenzel. "Augmented reality brings the real world into natural history dioramas with data visualizations and bioacoustics at the Carnegie Museum of Natural History". In: *Curator: The Museum Journal* 62.2 (2019), pp. 177–193. DOI: 10.1111/cura.12308
**SJR Q1**

5 Jacob Boesen Madsen, **Markus Tatzgern**, Claus B. Madsen, Dieter Schmalstieg, and Denis Kalkofen. "Temporal Coherence Strategies for Augmented Reality Labeling". In: *IEEE Transactions on Visualization and Computer Graphics (TVCG)* 22.4 (2016), pp. 1415–1423. DOI: 10.1109/TVCG.2016.2518318
**SJR Q1**, **CORE A***

6 **Markus Tatzgern**, Raphael Grasset, Eduardo Veas, Denis Kalkofen, Hartmut Seichter, and Dieter Schmalstieg. "Exploring Real World Points of Interest: Design and Evaluation of Object-Centric Exploration Techniques for Augmented Reality". In: *Pervasive and Mobile Computing* 18 (2015), pp. 55–70. DOI: 10.1016/j.pmcj.2014.08.010
**SJR Q1**

7 **Markus Tatzgern**, Denis Kalkofen, and Dieter Schmalstieg. "Multi-perspective compact explosion diagrams". In: *Computers & Graphics* 35.1 (2011), pp. 135–147. DOI: 10.1016/j.cag.2010.11.005

**Conference**

1 Irina Paraschivoiu, Thomas Layer-Wagner, Klaus Neundlinger, Simone Rack, and **Markus Tatzgern**. "Where's the Team Spirit? An Exploratory Study on Team Development Through Co-located Tablet-Based VR". in: *Proceedings of ACM Human-Computer Interaction (PACMHCI), Special Issue ACM Conference on Computer-Supported Cooperative Work and Social Computing (CSCW)* 10.6 (2026), pp. 1–31. DOI: 10.1145/3817040
**CORE A**

2 Lucchas Ribeiro Skreinig, Denis Kalkofen, David Ritter, Peter Mohr, **Markus Tatzgern**, Dieter Schmalstieg, and Ana Stanescu. "Magnification Lenses for Small-Scale Augmented Reality Instructions". In: *Proceedings of IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*. To appear. 2026
**CORE A***

3 Helen Stefanidi°, Jan-Hendrik Sünderkamp°, **Markus Tatzgern**, Alina Itzlinger, and Alexander Meschtscherjakov. "Everyday ARctivities in-the-Wild: Insights from a Farmers Market Field Study". In: *Proceedings of the Nordic Conference on Human-Computer Interaction (NordiCHI)*. ACM, 2026, pp. 1–11. DOI: 10.1145/3829807.3829863

4 Faith Young°, Anna Lienbacher, Janne Mascha Beuthel, Erich Prem, Clemens Havas, and **Markus Tatzgern**. "Ethics and AI: Designing an Interactive Self-Assessment Tool for Ethical Implementation of AI-enabled Projects". In: *Proceedings of the Nordic Conference on Human-Computer Interaction (NordiCHI)*. ACM, 2026, pp. 1–18. DOI: 10.1145/3829807.3829864

5 Jan-Hendrik Sünderkamp°, Helen Stefanidi°, Alexander Meschtscherjakov, and **Markus Tatzgern**. "From Lab to Sidewalk: Exploring AR Interaction Techniques Outdoors and in Motion using Fitts' Law". In: *Proceedings of ACM Symp. on Virtual Reality Software and Technology (VRST)*. 2025. DOI: 10.1145/3756884.3766006

6 Sebastian Hubenschmid, Johannes Zagermann, Robin Erb, Tiare Feuchtner, Jens Grubert, **Markus Tatzgern**, Dieter Schmalstieg, and Harald Reiterer. "SpatialMouse: A Hybrid Pointing Device for Seamless Interaction Across 2D and 3D Spaces". In: *Proceedings of ACM Symp. on Virtual Reality Software and Technology (VRST)*. 2025. DOI: 10.1145/3756884.3766047

7 Helen Stefanidi°, Jan-Hendrik Sünderkamp°, **Markus Tatzgern**, Alina Itzlinger, and Alexander Meschtscherjakov. "You're making things AR-kward: Exploring Augmented Reality In-the-Wild". In: *Proceedings of ACM Human-Computer Interaction (PACMHCI), Special Issue ACM International Conference on Mobile Human-Computer Interaction (MobileHCI)* 9.5 (2025), pp. 1–24. DOI: 10.1145/3743740
**Honorable Mention for Best Paper**

8 Jan Hendrik Plümer°, Kevin Yu, Ulrich Eck, Denis Kalkofen, Philipp Steininger, Nassir Navab, and **Markus Tatzgern**. "XR Prototyping of Mixed Reality Visualizations: Compensating Interaction Latency for a Medical Imaging Robot". In: *Proceedings of IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*. 2024. DOI: 10.1109/ISMAR62088.2024.00014
**CORE A***
**Honorable Mention for Best Student Paper**

9 Lucchas Ribeiro Skreinig, Peter Mohr, Blanca Berger, **Markus Tatzgern**, Dieter Schmalstieg, and Denis Kalkofen. "Immersive Authoring by Demonstration of Industrial Procedures". In: *Proceedings of IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*. 2024, pp. 1293–1302. DOI: 10.1109/ISMAR62088.2024.00146
**CORE A***

10 Helen Stefanidi°, **Markus Tatzgern**, and Alexander Meschtscherjakov. "Augmented Reality on the Move: A Systematic Literature Review for Vulnerable Road Users". In: *Proceedings of ACM Human-Computer Interaction (PACMHCI), Special Issue ACM International Conference on Mobile Human-Computer Interaction (MobileHCI)* 8 (2024), pp. 1–30. DOI: 10.1145/3676490

11 Jan Hendrik Plümer° and **Markus Tatzgern**. "Towards a Framework for Validating XR Prototyping for Performance Evaluations of Simulated User Experiences". In: *Proceedings of IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*. 2023, pp. 810–819. DOI: 10.1109/ISMAR59233.2023.00096
**CORE A***

12 Gerlinde Emsenhuber°, Tobias Langlotz, Denis Kalkofen, Jonathan Sutton, and **Markus Tatzgern**. "Eye-Perspective View Management for Optical See-Through Head-Mounted Displays". In: *Proceedings of ACM Conference on Human Factors in Computing Systems (CHI)*. New York, NY, USA: Association for Computing Machinery, 2023. DOI: 10.1145/3544548.3581059
**CORE A***

13 **Markus Tatzgern**, Michael Domhardt, Martin Wolf†, Michael Cenger, Gerlinde Emsenhuber, Radomir Dinic, Nathalie Gerner, and Arnulf Hartl. "AirRes Mask: A Precise and Robust Virtual Reality Breathing Interface Utilizing Breathing Resistance as Output Modality". In: *Proceedings of ACM Conference on Human Factors in Computing Systems (CHI)*. CHI '22. New York, NY, USA: Association for Computing Machinery, 2022. DOI: 10.1145/3491102.3502090
**CORE A***
**Honorable Mention for Best Paper**

14 **Markus Tatzgern** and Christoph Birgmann†. "Exploring Input Approximations for Control Panels in Virtual Reality". In: *Proceedings of IEEE Virtual Reality (VR)*. 2021. DOI: 10.1109/VR50410.2021.00092
**CORE A***

15 Masahiro Yamaguchi, Shohei Mori, Peter Mohr, **Markus Tatzgern**, Ana Stanescu, Hideo Saito, and Denis Kalkofen. "Video-Annotated Augmented Reality Assembly Tutorials". In: *Proceedings of ACM Symposium on User Interface Software and Technology (UIST)*. 2020, pp. 1010–1022. DOI: 10.1145/3379337.3415819
**CORE A***

16 Peter Mohr, **Markus Tatzgern**, Tobias Langlotz, Andreas Lang, Dieter Schmalstieg, and Denis Kalkofen. "TrackCap: Enabling Smartphones for 3D Interaction on Mobile Head-Mounted Displays". In: *Proceedings of ACM Conference on Human Factors in Computing Systems (CHI)*. 2019, pp. 1–11. DOI: 10.1145/3290605.3300815
**CORE A***

17 Peter Mohr, **Markus Tatzgern**, Jens Grubert, Dieter Schmalstieg, and Denis Kalkofen. "Adaptive User Perspective Rendering for Handheld Augmented Reality". In: *IEEE Symposium on 3D User Interfaces (3DUI)*. 2017, pp. 176–181. DOI: 10.1109/3DUI.2017.7893336
**Honorable Mention for Best Paper**

18 Peter Mohr, David Mandl, **Markus Tatzgern**, Eduardo E. Veas, Dieter Schmalstieg, and Denis Kalkofen. "Retargeting Video Tutorials Showing Tools With Surface Contact to Augmented Reality". In: *Proceedings of ACM Conference on Human Factors in Computing Systems (CHI)*. ACM, 2017, pp. 6547–6558. DOI: 10.1145/3025453.3025688
**CORE A***

19 **Markus Tatzgern**, Valeria Orso, Denis Kalkofen, Giulio Jacucci, Luciano Gamberini, and Dieter Schmalstieg. "Adaptive information density for augmented reality displays". In: *Proceedings of IEEE Virtual Reality (VR)*. 2016, pp. 83–92. DOI: 10.1109/VR.2016.7504691
**CORE A***

20 **Markus Tatzgern**, Denis Kalkofen, Raphael Grasset, and Dieter Schmalstieg. "Hedgehog Labeling: View Management Techniques for External Labels in 3D Space". In: *Proceedings of IEEE Virtual Reality (VR)*. 2014, pp. 1–6. DOI: 10.1109/VR.2014.6802046
**CORE A***

21 **Markus Tatzgern**, Raphael Grasset, Denis Kalkofen, and Dieter Schmalstieg. "Transitional Augmented Reality Navigation for Live Captured Scenes". In: *Proceedings of IEEE Virtual Reality (VR)*. 2014, pp. 21–26. DOI: 10.1109/VR.2014.6802045
**CORE A***

22 **Markus Tatzgern**, Denis Kalkofen, and Dieter Schmalstieg. "Dynamic compact visualizations for augmented reality". In: *Proceedings of IEEE Virtual Reality (VR)*. 2013. DOI: 10.1109/VR.2013.6549347
**CORE A***

23 **Markus Tatzgern**, Raphael Grasset, Eduardo Veas, Denis Kalkofen, Hartmut Seichter, and Dieter Schmalstieg. "Exploring Distant Objects with Augmented Reality". In: *Joint Virtual Reality Conference of EGVE - EuroVR*. ed. by Betty Mohler, Bruno Raffin, Hideo Saito, and Oliver Staadt. 2013, pp. 1–11. DOI: 10.2312/EGVE.JVRC13.049-056

24 Raphael Grasset, Tobias Langlotz, Denis Kalkofen, **Markus Tatzgern**, and Dieter Schmalstieg. "Image-Driven View Management for Augmented Reality Browsers". In: *Proceedings of IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*. 2012, pp. 177–186. DOI: 10.1109/ISMAR.2012.6402555
**CORE A***

25 **Markus Tatzgern**, Denis Kalkofen, and Dieter Schmalstieg. "Compact Explosion Diagrams". In: *Proceedings of the International Symposium on Non-Photorealistic Animation and Rendering*. 2010, pp. 17–26. DOI: 10.1145/1809939.1809942

26 Denis Kalkofen, **Markus Tatzgern**, and Dieter Schmalstieg. "Explosion Diagrams in Augmented Reality". In: *Proceedings of IEEE Virtual Reality (VR)*. 2009, pp. 71–78. DOI: 10.1109/VR.2009.4811001
**CORE A***

**Posters, Workshops, Demos and Other Publications**

1 Lucchas Ribeiro Skreinig, Ana Stanescu, Peter Mohr, Christoph Ebner, **Markus Tatzgern**, Dieter Schmalstieg, and Denis Kalkofen. "Augmented Reality Action Summaries for Situated Instructions of Repetitive Tasks". In: *IEEE International Symposium on Mixed and Augmented Reality Adjunct (ISMAR-Adjunct)*. Poster, to appear. 2026

2 Faith Young°, **Markus Tatzgern**, Alexander Meschtscherjakov, and Jan David Smeddinck. "Conceptualising an Initial Design Space for Guidance in Digital Physical Activity Support". In: *Adjunct Proceedings of the Nordic Conference on Human-Computer Interaction (NordiCHI Adjunct)*. Poster. ACM, 2026, pp. 1–9. DOI: 10.1145/3821402.3830134

3 Helen Stefanidi°, Vincent van Rheden, Linjia He, Michelle Adiwangsa, Jan-Hendrik Sünderkamp°, Alina Itzlinger, Andrii Matviienko, Julie R. Williamson, Ian Oakley, **Markus Tatzgern**, and Alexander Meschtscherjakov. "Next Steps for Augmented Reality On-the-Move: Challenges and Opportunities". In: *Proceedings of the Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA)*. CHI EA '26. ACM, 2026. DOI: 10.1145/3772363.3778775

4 Sebastian Hubenschmid, Johannes Zagermann, Niklas Elmqvist, Tiare Feuchtner, Jens Grubert, **Markus Tatzgern**, Dieter Schmalstieg, and Harald Reiterer. "Revisiting Hybrid Input Devices for Immersive Analytics". In: *IEEE Conference on Human Factors in Immersive Analytics (HFIA), IEEE VIS Workshop*. 2025, pp. 29–33. DOI: 10.1109/HFIA68651.2025.00011

5 Gerlinde Emsenhuber°, Michael Smirnov, Tobias Langlotz, Denis Kalkofen, Stefanie Zollmann, and **Markus Tatzgern**. "Mobile Eye-perspective Rendering: Alignment of Image-based Vision Augmentations for Optical See-through Head-mounted Displays". In: *IEEE International Symposium on Mixed and Augmented Reality Adjunct (ISMAR-Adjunct)*. 2025, pp. 917–918. DOI: 10.1109/ISMAR-Adjunct68609.2025.00250

6 Helen Stefanidi°, Jan-Hendrik Sünderkamp°, Alina Itzlinger, **Markus Tatzgern**, and Alexander Meschtscherjakov. "Exploring AR In-the-Wild: An Autoethnographic Study at a Christmas Market". In: *Companion Publication of the 2025 ACM Designing Interactive Systems Conference (DIS)*. 2025, p. 7. DOI: 10.1145/3715668.3736368

7 Helen Stefanidi°, **Markus Tatzgern**, Alina Itzlinger, and Alexander Meschtscherjakov. "Bringing Adaptive On-the-Move AR to Outdoor Environments". In: *ACM Conference on Human Factors in Computing Systems (CHI), Workshop Artificial Intelligence and Human Interfaces*. 2025

8 **Markus Tatzgern**, David Fasching, Vivian Gómez, Sebastián Calvache, and Pablo Figueroa. "Passive Haptic Feedback for Control Elements in Virtual Reality". In: *IEEE Virtual Reality and 3D User Interfaces - Abstracts and Workshops (VRW)*. 2024, pp. 749–750. DOI: 10.1109/VRW62533.2024.00171

9 Gerlinde Emsenhuber°, Michael Domhardt, Tobias Langlotz, Denis Kalkofen, and **Markus Tatzgern**. "Towards Eye-Perspective Rendering for Optical See-Through Head-Mounted Displays". In: *IEEE Virtual Reality and 3D User Interfaces - Abstracts and Workshops (VRW)*. 2022, pp. 640–641. DOI: 10.1109/VRW55335.2022.00171

10 Denis Kalkofen, Shohei Mori, and **Markus Tatzgern**. "Visualization and Graphics in Mixed Reality". In: *Eurographics - Tutorials*. The Eurographics Association, 2021. DOI: 10.2312/egt.20211036

11 **Markus Tatzgern**, Denis Kalkofen, Raphael Grasset, and Dieter Schmalstieg. "Embedded Virtual Views for Augmented Reality Navigation". In: *Workshop on Visualization in Mixed Reality Environments, International Symposium on Mixed and Augmented Reality (ISMAR)*. 2011. DOI: 10.13140/2.1.4398.1769

## Patents

2026 Virtual Content Adjustment. Gerlinde Emsenhuber, Markus Tatzgern. US Patent Application, Publication number US 2026/0260439 A1, September 2026 (applicant: Qualcomm Incorporated; filed March 2025)

2014 Hierarchical clustering for view management augmented reality. Markus Tatzgern, Denis Kalkofen, Dieter Schmalstieg, Raphael Grasset. US Patent, Publication number 14/640,981, 2014

2013 Image-Driven View Management for Annotations. Raphael Grasset, Markus Tatzgern, Tobias Langlotz, Denis Kalkofen and Dieter Schmalstieg. WO Patent, Publication number 2013176830, November 2013

2012 Efficient Information Presentation for Augmented Reality. Markus Tatzgern, Denis Kalkofen and Dieter Schmalstieg. WO Patent, Publication number 2012033768, March 2012

## Supervision

Official supervision and co-supervision of five PhD students, four of which are located at Salzburg University of Applied Sciences.

ongoing Supervision of PhD thesis by Gerlinde Emsenhuber,
*Eye-perspective Rendering for Optical See-through Head-mounted Displays*
Located at Salzburg University of Applied Sciences, PhD programme at Graz University of Technology with Prof. Denis Kalkofen

ongoing Supervision of PhD thesis by Jan Hendrik Plümer,
*Validation of XR Prototyping for Human-robot Interaction*
Located at Salzburg University of Applied Sciences, PhD programme at Graz University of Technology with Prof. Denis Kalkofen

ongoing Supervision of PhD thesis by Jan-Hendrik Sünderkamp,
*Effective XR Interaction while Being on the Move*
Located at Salzburg University of Applied Sciences, PhD programme at University of Salzburg with Prof. Alexander Meschtscherjakov

ongoing Co-supervision of PhD thesis by Faith Young,
*Ethical AI-based Health Interventions*
Located at Salzburg University of Applied Sciences, PhD programme at University of Salzburg with Prof. Alexander Meschtscherjakov

ongoing Co-supervision of PhD thesis by Helen (Eleni) Stefanidi,
*XR User Experiences in Outdoor Environments*
PhD programme at University of Salzburg with Prof. Alexander Meschtscherjakov

Regular supervision of Bachelor's theses (53 completed) and Master's theses (24 completed).

## Teaching

### Salzburg University of Applied Sciences, Bachelor Level

2015–2026 Game Development 1 (Lecture and Lab)
2016–2020 Game Development 2 (Lecture and Lab)
2016–2021 Computer Graphics (Lecture and Lab)
2016 Advanced Topics of Game Development and Augmented Reality (Lecture and Lab)
2016–2019 Study & Presentation Skills (Scientific Writing)
2020–2026 Seminar Paper Computer Science (Scientific Writing)
2016–2026 Bachelor Thesis Seminar
2024–2026 Augmented & Virtual Reality (Lecture and Lab)

### Salzburg University of Applied Sciences, Master Level

2017–2026 Mixed Reality Technologies (Lecture and Lab)
2016 Mixed Reality Interaction (Lecture and Lab)
2017–2024 Real-Time Visualization (Lecture and Lab)
2020, 2025 Interaction Approaches & Technologies (Lecture and Lab)
2023 Applied Games (Lecture and Lab)
2024 Guest Lecture Game & Simulation Engineering (Lecture and Lab)
2019–2023 Research Seminar
2024–2026 Master Thesis Seminar 1
2024–2026 Research Method and Study Design
2016–2026 Master Thesis Seminar 2

## Professional Development

10/2025 Gender and Diversity Training
1 half-day course

04/2023–06/2023 Leadership Training
Seminar series covering 2.5 full-day modules.

09/2016 Higher Education Didactics: Instructional Design
1 half-day module

10/2015 Higher Education Didactics: Definition and Assessment of Learning Outcomes
1 full-day module

10/2015 Higher Education Didactics: Supervision of Thesis Work
1 full-day module

10/2015 Higher Education Didactics: Reflective Learning. Stimulating Conscious Learning Processes
1 half-day module

09/2015 Higher Education Didactics: Didactics of an Online Course
1 full-day module

09/2015 Higher Education Didactics: Exams and Skill Acquisition
1 full-day module

12/2014 Protecting Intellectual Property: Patent Applications and Exploitation
1 full-day module

11/2014 EU Project Acquisition and Management
1 full-day module

## Awards and Honours

2025 Honorable Mention for Best Paper at the ACM International Conference on Mobile Human-Computer Interaction (MobileHCI), Paper: You're making things AR-kward: Exploring Augmented Reality In-the-Wild

2024 Honorable Mention for Best Student Paper at the IEEE Symposium for Mixed and Augmented Reality (ISMAR) (**CORE A***), Paper: XR Prototyping of Mixed Reality Visualizations: Compensating Interaction Latency for a Medical Imaging Robot

2022 Honorable Mention for Best Paper at the ACM CHI Conference on Human Factors in Computing Systems (CHI) (**CORE A***), Paper: AirRes Mask: A Precise and Robust Virtual Reality Breathing Interface Utilizing Breathing Resistance as Output Modality

2017 Honorable Mention for Best Paper at the IEEE Symposium on 3D User Interfaces (3DUI), Paper: Adaptive User Perspective Rendering for Handheld Augmented Reality

2011, 2013, 2015 Distinct Inventor at Graz University of Technology

## Editorial and Conference Leadership

### Chair Duties

2025 IEEE International Symposium on Mixed and Augmented Reality (ISMAR), Daejeon, South Korea, Science and Technology Papers Chair for Conference and Journal Track (transition to single track ISMAR)

2024 IEEE International Symposium on Mixed and Augmented Reality (ISMAR), Greater Seattle Area, Science and Technology Program Chair for Conference and Journal Track

2024, 2025 ACM Conference on Human Factors in Computing Systems (CHI), Associate Chair, Subcommittee User Experience and Usability

2023 IEEE International Symposium on Mixed and Augmented Reality (ISMAR), Sydney, Poster Chair

### Editorial Duties

since 2026 Associate Editor of IEEE Transactions on Visualization and Computer Graphics

since 2024 Editorial Board of Springer Journal Virtual Reality

since 2021 Editorial Board of Journal Virtual Reality & Intelligent Hardware

since 2020 Editorial Board of Augmented Reality: Review Editor for Frontiers in Virtual Reality

## Service

### Committee Duties

2026 IEEE International Symposium on Mixed and Augmented Reality (ISMAR)

2026 European Association for eXtended Reality (EuroXR) International Conference

2026 IEEE Visualization and Graphics Technical Community (VGTC) Virtual Reality Best Dissertation Award Committee Member

2023, 2025, 2027 IEEE Virtual Reality and 3D User Interfaces (VR), conference and journal track (IEEE Transactions on Visualization and Computer Graphics)

2022 IEEE International Conference on Artificial Intelligence & Virtual Reality (AIVR)

2022 IEEE Virtual Reality and 3D User Interfaces (VR), journal track (IEEE Transactions on Visualization and Computer Graphics)

2021 IEEE International Symposium on Mixed and Augmented Reality (ISMAR), journal track (IEEE Transactions on Visualization and Computer Graphics)

2020 IEEE International Symposium on Mixed and Augmented Reality (ISMAR), conference track

2016, 2019–2022 IEEE Virtual Reality and 3D User Interfaces (VR)

2019 ACM Symposium on Virtual Reality Software and Technology (VRST)

2018 FH Forschungsforum, Chair: Digitized - Transformation durch Augmented und Virtual Reality

### Other Roles

2026 AVRiL 2026, Committee Member for Competition XR Learning for German Informatics Society (GI)

since 10/2025 Chapter Officer (Treasurer) of Austrian ACM Special Interest Group on Computer-Human Interaction (SIGCHI) Chapter

2025 Founding Member of Forum New Media Austria (FNMA) Special Interest Group Immersive Education

2025, 2026 EdTech Austria Fellowship Program - Mentor

since 04/2025 Member of advisory board in FFG funded research project GreenTouch - XR in Ergotherapy, Austria

since 10/2023 Scientific Collaboration and Consulting in Digital Health Technologies, Ludwig Boltzmann Institute - Digital Health and Prevention, Austria

since 2023 Founding Member of Austrian ACM Special Interest Group on Computer-Human Interaction (SIGCHI) Chapter

2023 Jury Member for the Content Vienna Award funded by the Vienna Business Agency, Austria

2022 Expert Evaluator for Accreditation of Study Program for University of Applied Management, Germany

2020, 2021, 2026 Doctoral Consortium Mentor at IEEE International Symposium on Mixed and Augmented Reality (ISMAR)

2017, 2018 Expert Evaluator for Austrian Research Promotion Agency (FFG)

### Reviewing

since 2009 IEEE International Symposium on Mixed and Augmented Reality (ISMAR)

since 2011 IEEE Virtual Reality (VR)

since 2011 Computers & Graphics (C&G)

since 2013 ACM Conference on Human Factors in Computing Systems (CHI)

since 2017 IEEE Transactions on Visualization and Computer Graphics

since 2016 ACM Symposium on Virtual Reality Software and Technology (VRST)

2025 Springer Journal Virtual Reality

2023 International Conference on Human-Computer Interaction (INTERACT)

2022, 2023, 2026 Doctoral Consortium Reviewer at IEEE International Symposium on Mixed and Augmented Reality (ISMAR)

2020 ACM Symposium on User Interface Software and Technology (UIST)

2022 IEEE Computer Graphics and Applications (CG&A)

2012 Eurographics

2012 ACM International Conference on Human-Computer Interaction with Mobile Devices and Services (MobileHCI)

2015 ACM Symposium on Spatial User Interaction (SUI)

2015 International Conference on Mobile and Ubiquitous Multimedia (MUM)

2016 Journal of Virtual Reality and Broadcasting (JVRB)

2020 ACM Interaction Design and Children (IDC)

### Reviewing Recognitions

2025, 2026 Special Recognition for Outstanding Reviews for ACM CHI Conference on Human Factors in Computing Systems (CHI), Papers Track

2024 Reviewer Certificate for reviewing efforts from IEEE Transactions on Visualization and Computer Graphics (TVCG)

2020 Special Recognition for Outstanding Reviews for ACM Symposium on User Interface Software and Technology (UIST), Papers Track

2020 Special Recognition for Outstanding Reviews for IEEE Symposium for Mixed and Augmented Reality, Papers Track

2019, 2021, 2022 Recognition for 'highly useful' reviews for IEEE Symposium for Mixed and Augmented Reality, Journal Papers Track

## Invited Talks and Presentations

2026 Invited talk, “Everyday XR - Moving Out of the Lab”, Wintergraph 2026, 19th Visual Computing Winter Workshop, Austria

2025 Panel discussion, “Immersion x Education for Youth”, culttech summit, Vienna

2024 Invited talk, “EdTech - Augmented Learning Lab”, EdTech Summit, Salzburg

2022 Invited talk and panel discussion, “Trends and Potentials of EdTech”, EdTech Austria Roadshow

2021 Invited talk, “Supporting Primary School Education with Technology”, EdTech Austria - Round Table

2021 Tutorial, “Visualization and Graphics in Mixed Reality”, Eurographics 2021

2018 Invited talk, “Digitalization in Architecture and Construction - Use Cases of Virtual and Augmented Reality”, Brennpunkt Alpines Bauen, Austria

2018 Invited talk, “Virtual and Augmented Reality - Technological Maturity and Applications in Tourism”, Brennpunkt eTourism, Austria

2018 Invited talk, “State-of-the-Art of Virtual and Augmented Reality Technology and Applications”, Austrian Armed Forces, Austria

2018 Invited talk, “Virtual Reality in E-commerce”, Marketing Club Salzburg, Austria

2018 Invited talk, “Augmented and Virtual Reality for Medical Education and Health Care”, 1. Salzburger Forum für Medizin und Digitalisierung, Paracelsus Medical University, Salzburg, Austria

2017 Invited talk, “Augmented and Virtual Reality: A building block of industry 4.0”, Alles Holz 4.0 oder was, Holzcluster, Salzburg, Austria

2017 Invited talk, “Virtual and Augmented Reality in Education”, eEducation Praxistagung, Pädagogische Hochschule Salzburg, Salzburg, Austria

2015 Invited talk, “Augmented Reality Technology and Applications”, Center for Human-Computer Interaction, University of Salzburg, Salzburg, Austria

2015 Invited talk, “Next Generation Augmented Reality Games”, Extended Games Day, Hochschule der Medien, Stuttgart, Germany

## Public Engagement (Selected Media)

2025 “Artificial, intelligent and educational?” / “Künstlich, intelligent – und bildend”, Salzburger Nachrichten (print, online), 16 December, Austria

2025 “Education in Salzburg making headway for using AI for teaching” / “Salzburger Bildung macht sich fit für KI in der Lehre”, Salzburger Nachrichten (print, online), 10 November, Austria

2024 “FH Salzburg presents new research strategy”, MyScience (online), 6 June

2023 “Saving time and money utilizing virtual prototyping” / “Mit virtuellen Prototypen Zeit und Kosten sparen” , Salzburger Nachrichten (print), 20 October, Austria

2022 “Introducing ’AirRes Mask’ that can introduce the concept of ’breathing’ into VR”, Gigazine (online), 16 May, Japan

2022 “This mask makes breathing in virtual reality more realistic”, edition.cnn.com / CNN Business (online), 15 May

2022 “Virtual reality mask adds realism by making it harder to breath”, NewScientist, newscientist.com, 11 May

2021 “The potential of Augmented Reality” / “Wie viel Potenzial steckt in Augmented Reality”, doIT digitale Karriere-Perspektiven, Der Standard (print), 27 November, Austria

2021 “Learning with digital support” / “Lernen gelingt mit digitalen Helfern”, Salzburger Nachrichten (print), 24. April, Austria

2021 “Fannys Friday: Game Science”, ORF / Austrian Broadcasting Corporation (television) - tvthek.orf.at, 8 January, Austria

2020 “Teaching Chemistry in a Virtual Laboratory”/“Chemie vermitteln lernen im virtuellen Labor”, Die Presse (print) - diepresse.com, 12 December, Austria

2019 “Augmented Reality in the Work Place”/ TV report “Austrian Broadcasting Corporation” (“Österreichischer Rundfunk”), 4 March, Austria

2018 “New Carnegie Museum app shows threat to wildflower diversity in the woods”, Pittsburgh Post-Gazette / post-gazette.com, 26 December, USA

2018 “Augmented Reality App Shows How Deer Overpopulation Affects Western PA’s Forests”, Pittsburgh’s NPR News Station / wesa.fm, 19 December, USA